Chapter 2.2: Quantile-Quantile plots

Q-Q Plots

We wish to investigate whether observations $y_1, \ldots, y_n$ may reasonably be regarded as a random sample from some theoretical distribution.

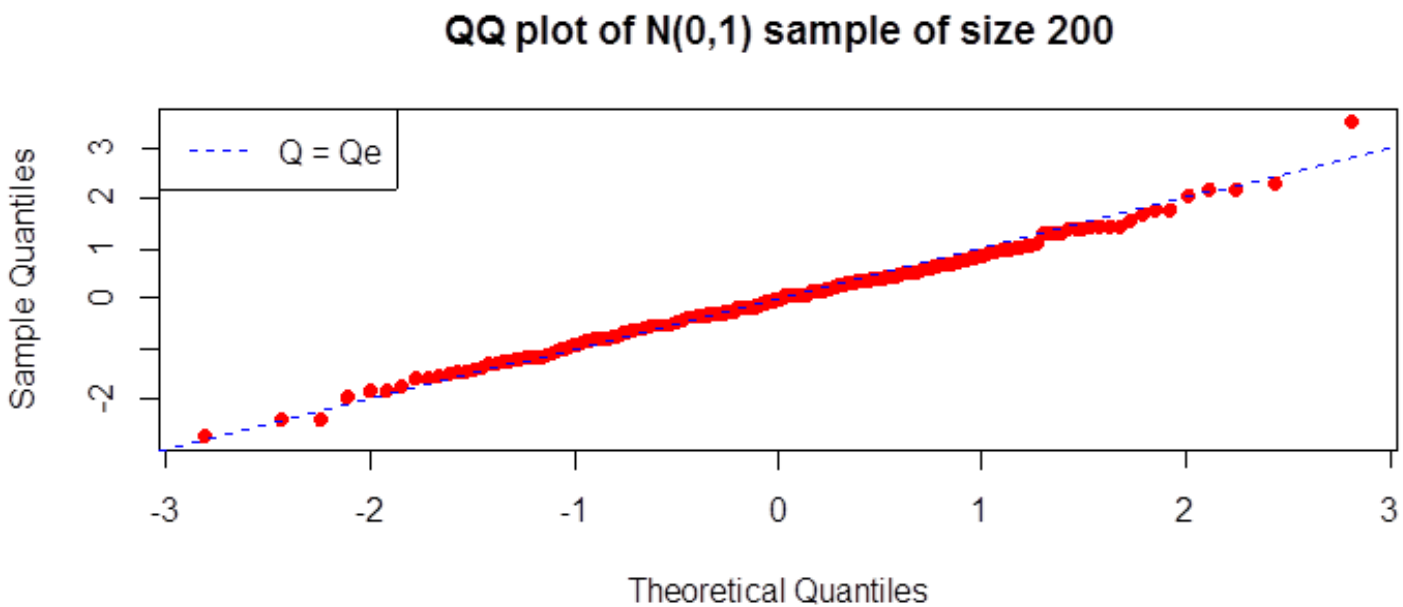
A Q-Q plot compares the *empirical* quantile function Q_e of the data with the *theoretical* quantile function Q of the distribution.

We plot $y_{(i)} (= Q_e(p_i))$ (on the y axis) against $Q(p_i)$ (on the x axis) for the probability points p_1, p_2 etc. as defined earlier.

If the points on the graph form an approximately straight line with slope 1 and intercept 0, the theoretical distribution is said to be compatible with the observed values, i.e.

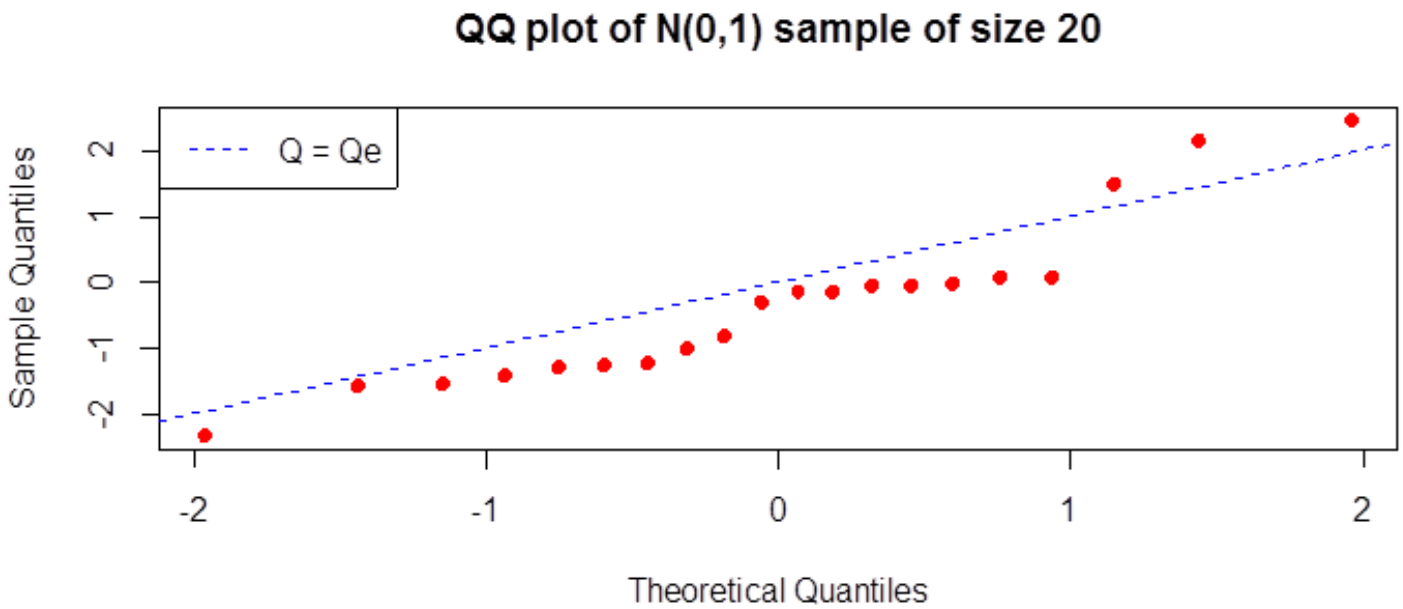
$$y_{(i)} \approx Q(p_i)$$

For example, data simulated from $N(0, 1)$ distn:



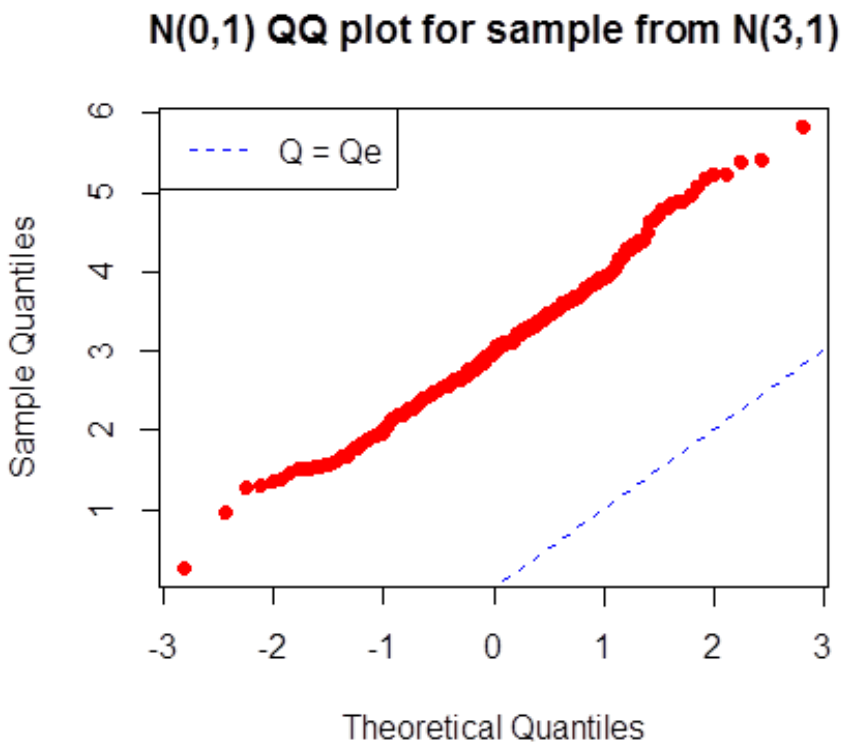
But, be careful when data set is small –plot will be far from perfect.

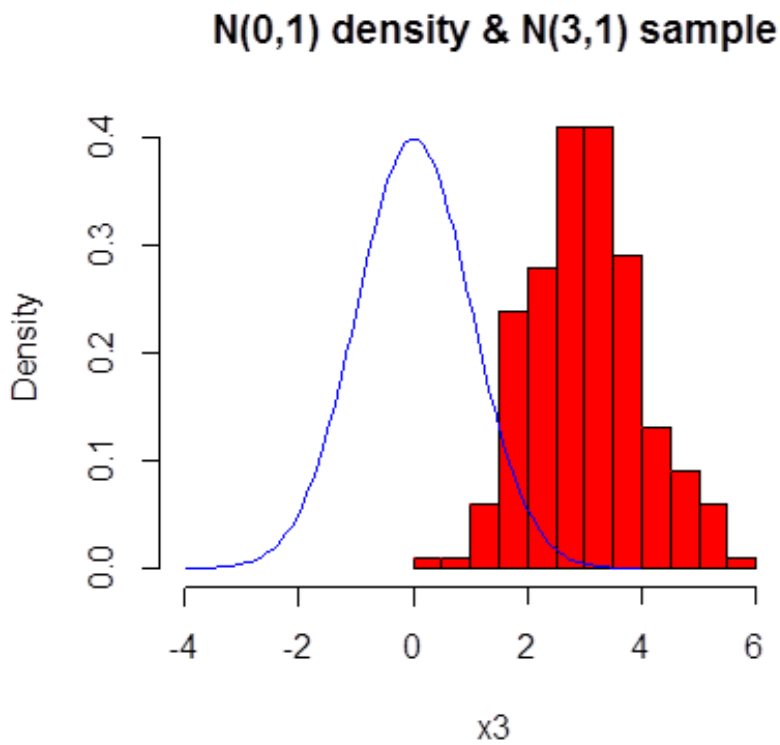
E.g. with $n = 20$ observations from $N(0, 1)$ distn:



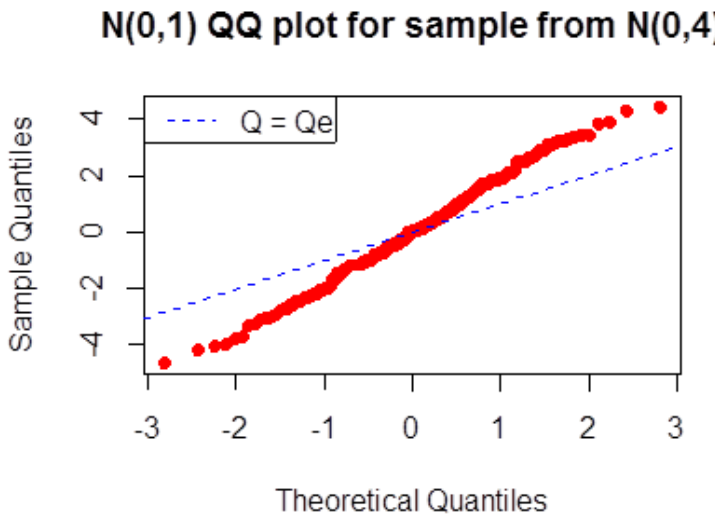
Q-Q plots can be used to identify how two distributions (e.g. data, theoretical) differ in location, spread or shape.

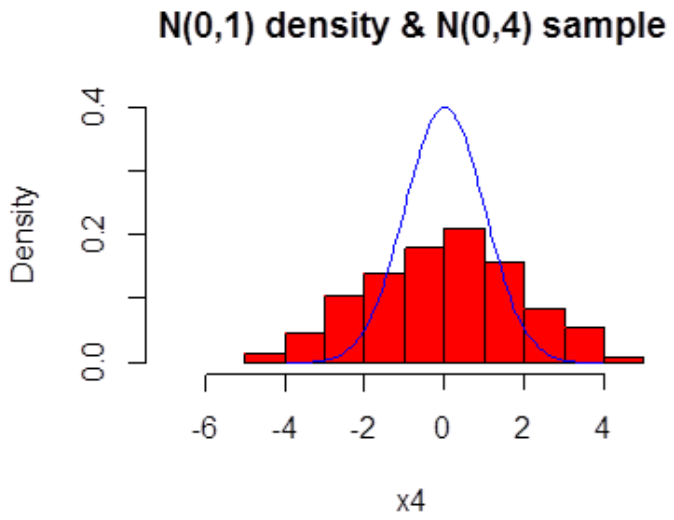
E.g.different location:



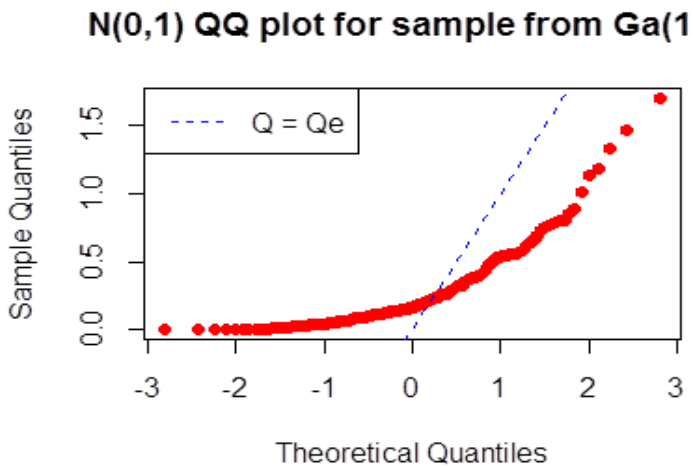


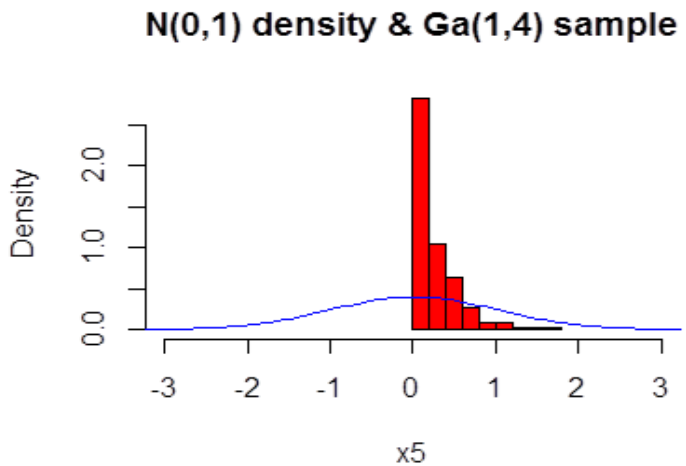
Different spread (same location):



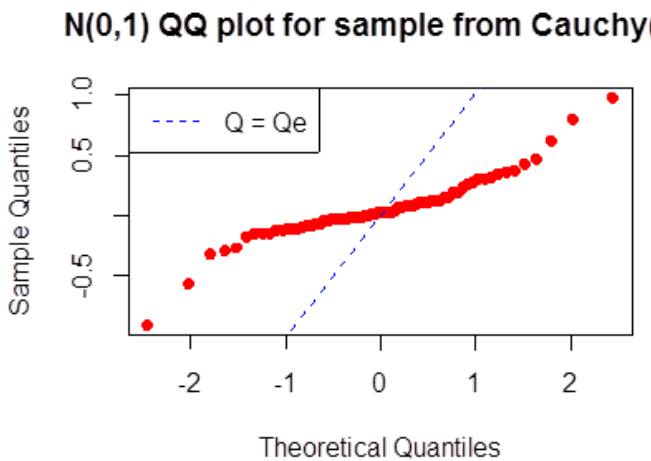


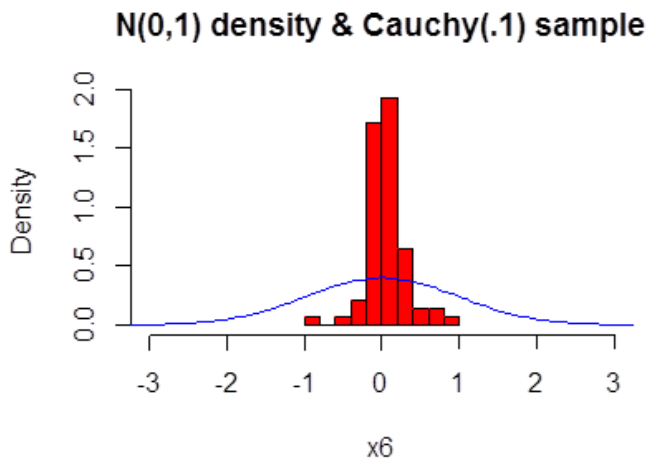
Different shape:



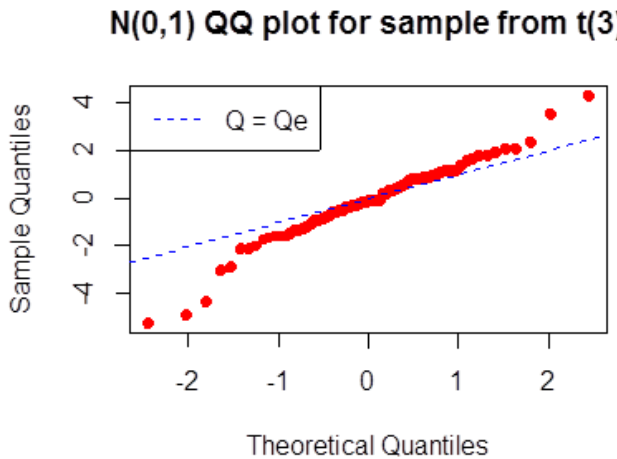


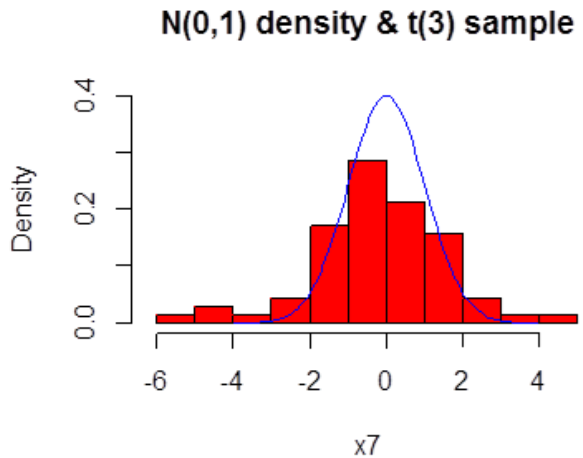
Short tails compared to normal distn:



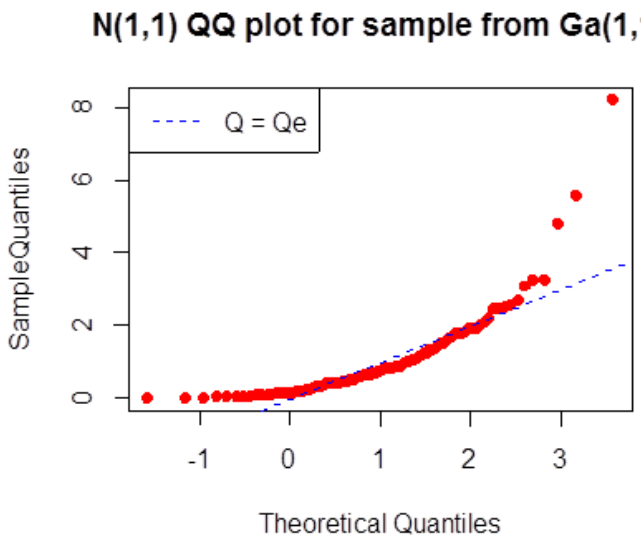


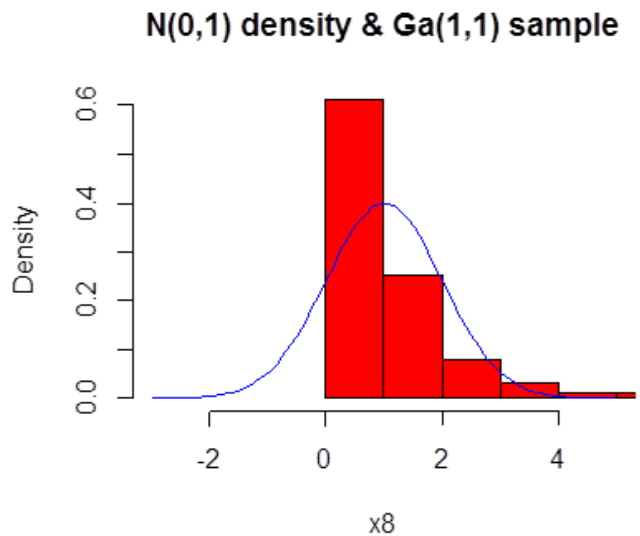
Long tails compared to normal distn:



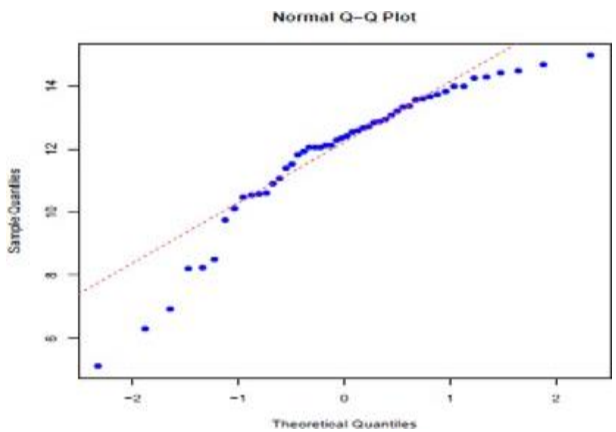


Positive skewness:





Negative skewness:



Refinement

If the data do not lie on the specific line $y_{(i)} = Q(p_i)$, but instead lie on a line with slope b and intercept a,

$$\text{i.e.} \quad y_{(i)} \approx a + bQ(p_i),$$

this indicates compatibility with the appropriate distribution scaled by b and shifted by a.

Sometimes (not always!) the linear transformation allows us to identify (i.e. give the name of) the distribution.

For example if Q is the quantile function of the $N(0, 1)$ distribution, then the relation suggests compatibility with the $N(a, b^2)$ distribution.

Example 2.1 (cont.)

Stamford Ozone Concentrations

We check for normality.

A normal Q-Q plot compares empirical quantiles of the data with the quantiles of the $N(0, 1)$ distribution.

The R function qqnorm constructs the plot.

R Console

```
R : Copyright 2003, The R Development Core Team
Version 1.6.2  (2003-01-10)

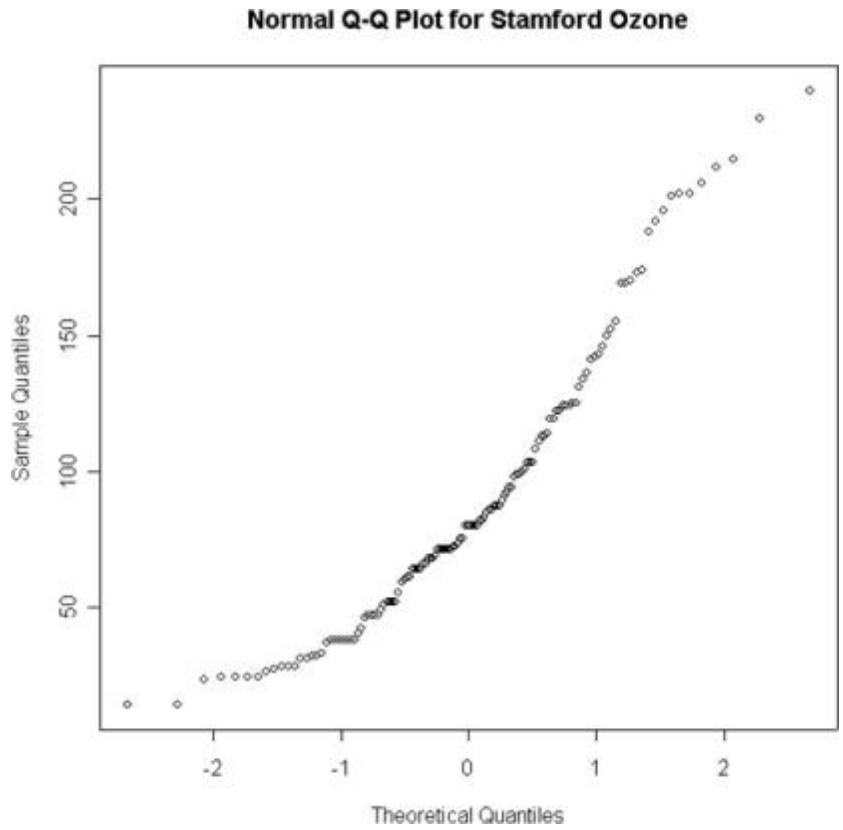
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type `license()' or `licence()' for distribution details.

R is a collaborative project with many contributors.
Type `contributors()' for more information.

Type `demo()' for some demos, `help()' for on-line help, or
`help.start()' for a HTML browser interface to help.
Type `q()' to quit R.

[Previously saved workspace restored]

> ozone=dget("ozone.R")
> attach(ozone)
> p=seq(0,1,0.1)
> quantile(stamford,probs=p)
   0%   10%   20%   30%   40%   50%   60%   70%   80%   90%  100%
 14.0  31.1  42.8  59.3  69.8  80.0  88.2 106.5 125.0 169.9 240.0
> qqnorm(stamford,main="Normal Q-Q Plot for Stamford Ozone")
>
```



- Is there a straight line that (approximately) goes through the points?
- In any case, that line would not go through zero, so intercept $\neq 0$.
- Also, slope $\neq 1$.

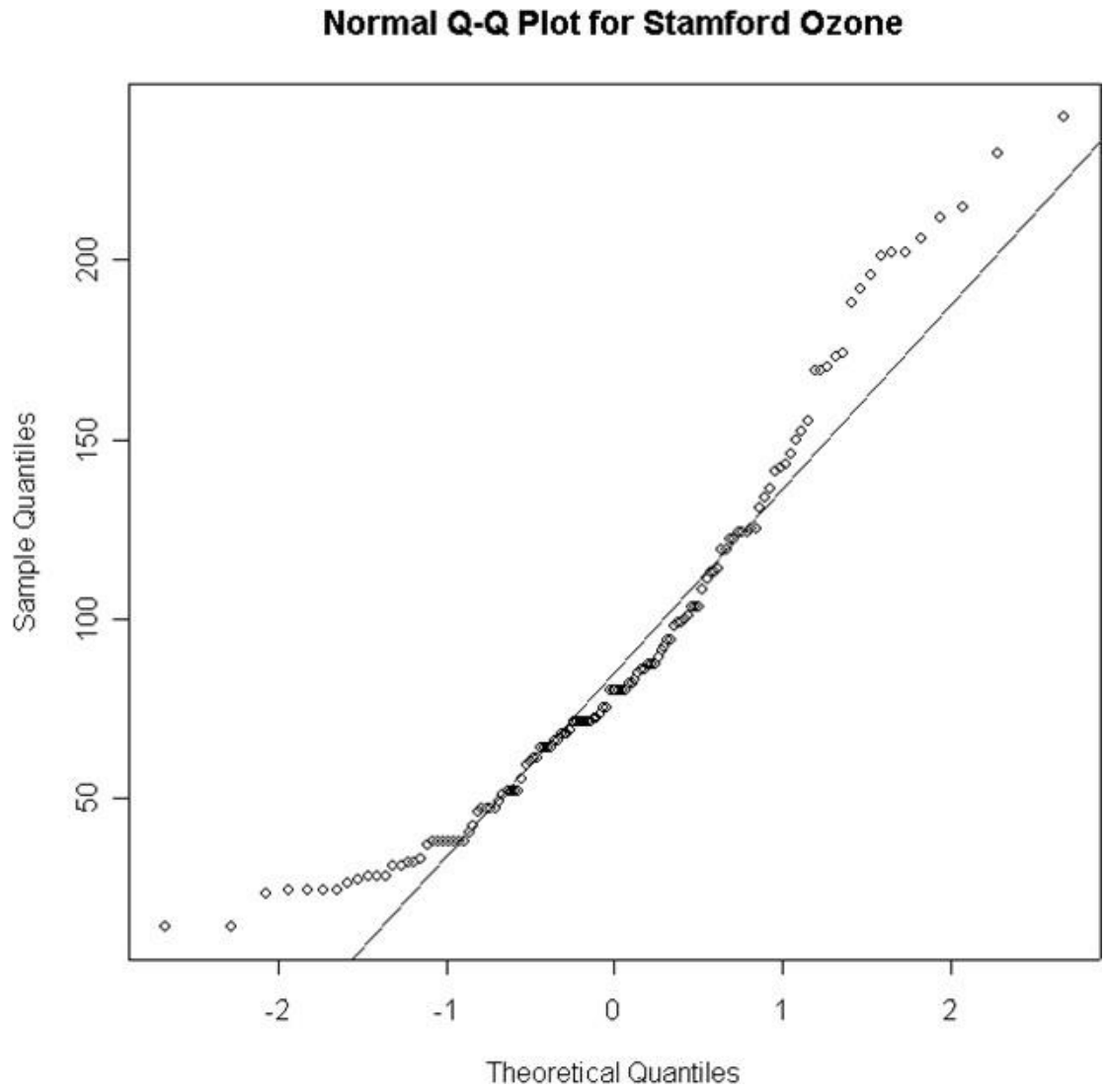
The function qqline adds the straight line

$$y = a + bx$$

corresponding to the normal distribution $N(a,\ b^2)$ whose lower and upper quartiles match those of the data (see later).

This is a resistant fit, unaffected by one or two extreme observations.

```
<environment: namespace:base>
> qqnorm(stamford,main="Normal Q-Q Plot for Stamford Ozone")
> qqline(stamford)
>
```



Plot indicates positive skewness.

The lack of linearity in the plotted quantiles suggests that the data are not well modelled by a normal distribution. Rather, the convex shape indicates **positive skewness** (as was seen in the histogram).

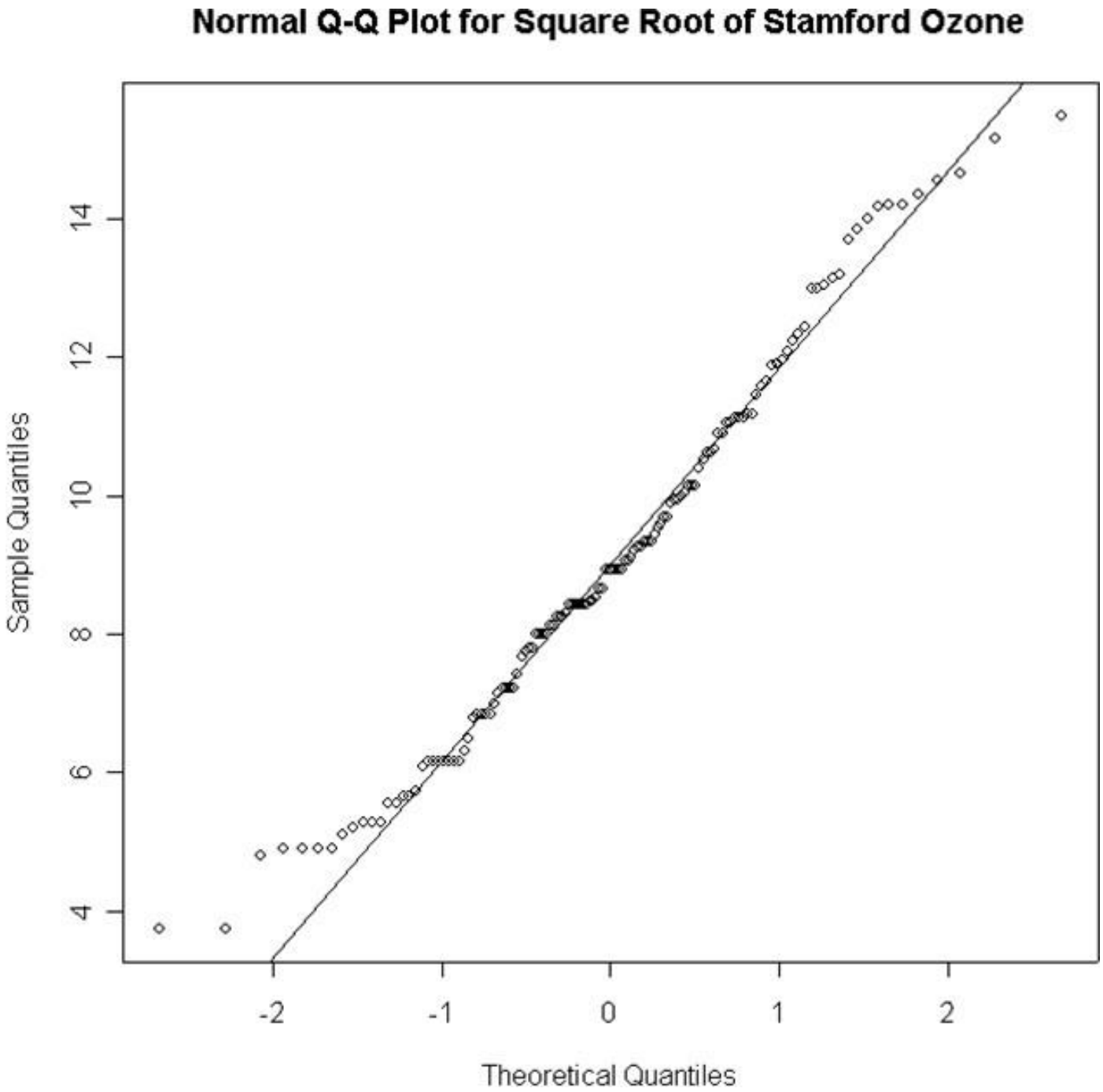
We investigate whether a transformation would help. A check is carried out as to whether the **square roots** of the Stamford ozone concentrations can be modelled by a normal distribution.

```
>
>
>
> qqnorm(stamford^0.5,main="Normal Q-Q Plot for Square Root of Stamford Ozone")
> qqline(stamford^0.5)
>
```

Notice that, when using R,

$$x \hat{} y$$

gives the yth power of x, and, of course, a power of 0.5 gives a square root.


Normal Q-Q Plot for Square Root of Stamford Ozone
Sample Quantiles
4
6
8
10
12
14
-2
-1
0
1
2
Theoretical Quantiles

The qqline suggests that the quantiles are reasonably well fitted by the straight line which has slope 2.844 and intercept 9.0 (see derivation on next slide).

Thus the square root of data are reasonably well modelled by the $N(9,\ 2.844^2)$ distribution.

Note that the sample mean and standard deviation are 9.1 and 2.715 respectively.

The equation of the straight line is obtained (fitted to the data) by using the first and third quartiles to determine two points on the line, with co-ordinates

$$(x_1, y_1) \text{ and } (x_2, y_2)$$

given by

$$(Q(0.25),\ Q_e(0.25)) \text{ and } (Q(0.75),\ Q_e(0.75))$$

Appropriate R commands give these as

$$(-0.675,\ 7.106) \text{ and } (0.675, 10.940)$$

Therefore:

Slope of line: $b = \frac{y_2 - y_1}{x_2 - x_1} = 2.844$

Intercept: $y_1 = a + bx_1 \Rightarrow \quad a = 9.023.$

So the line on the graph is

$$Q_e(p_i) = 9.023 + 2.844Q(p_i)$$

implying that $\sqrt{data}$ follow $N(9.023,\ 2.844^2)$.